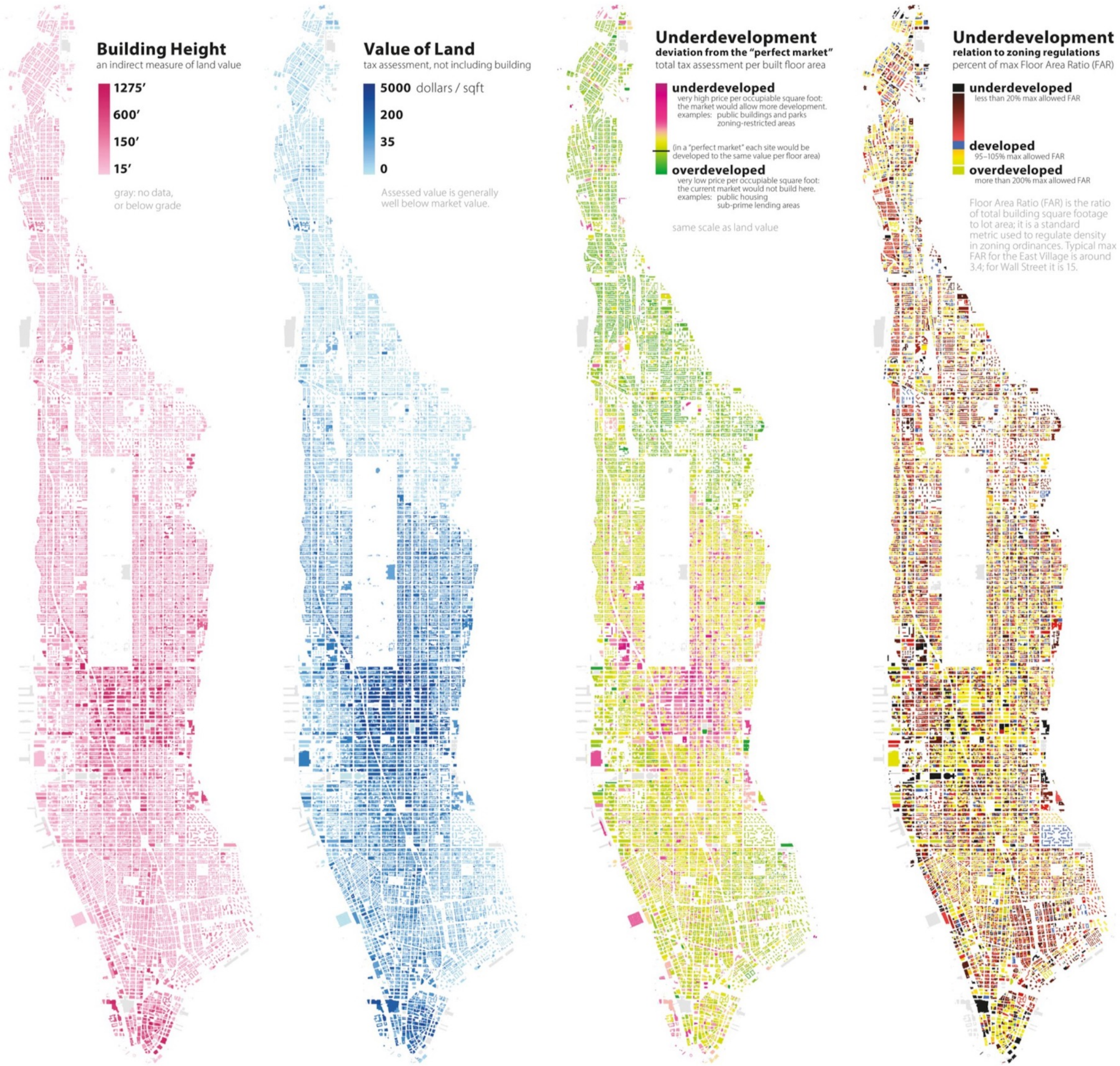
Map Spotlight 8: Color



Rankin, 2025

Process of Map Design

- Step 1:
  - Combine imagination and creativity to think about possible design methods, involving topics such as map purpose, format, data, and reference mapping techniques.

- Step 2:
  - Formulate a specific graphic plan. Consider different design strategies within the scope set in the first stage, such as the types of symbols, grading restrictions, lines, and colors, etc.

- Step 3:
  - The detailed plan of the map production includes all drawing methods and procedures, mainly the design, weight, color, and note size of the symbols.

Visual Impact (視覺影響)

- Maps show the geographic phenomenon through symbols.

- The basis of map interpretation comes from the effect of map symbols on sensory difference, including two important factors:
  - Graphic elements (圖形元素)
  - Visual variables (視覺變數)

Graphic Elements (圖形元素)

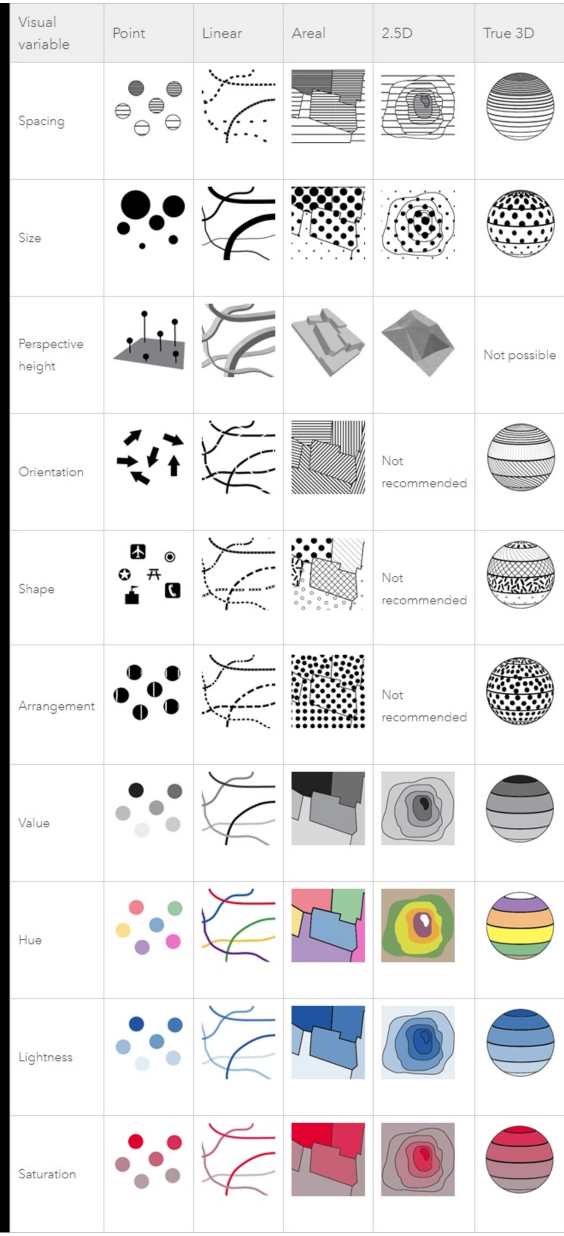
- Determine the appropriate dimension of the corresponding symbol according to the geographical phenomenon

- Symbol types and application:
  - Dot symbol
    - It is usually used to express the attributes of a specific location.
  - Line symbol
    - It is usually used to represent linear objects, such as roads, river, etc.
  - Area symbol
    - Have a specific area that has the same distribution of attribute values within the area.
  - Volume symbol
    - Show the distribution of height or intensity in a geographic phenomenon.

Visual Variables (視覺變數)

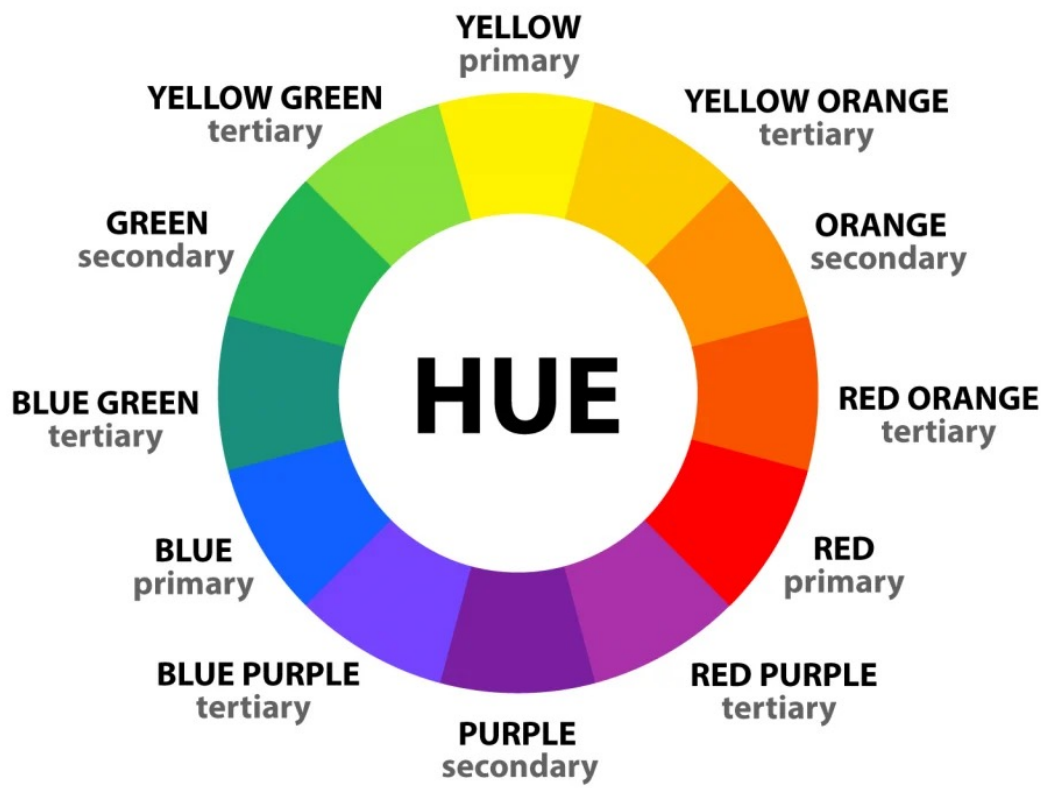
- The use of visual variables causes visual differences in various graphic elements, and represents different types or degrees of phenomena in map design.

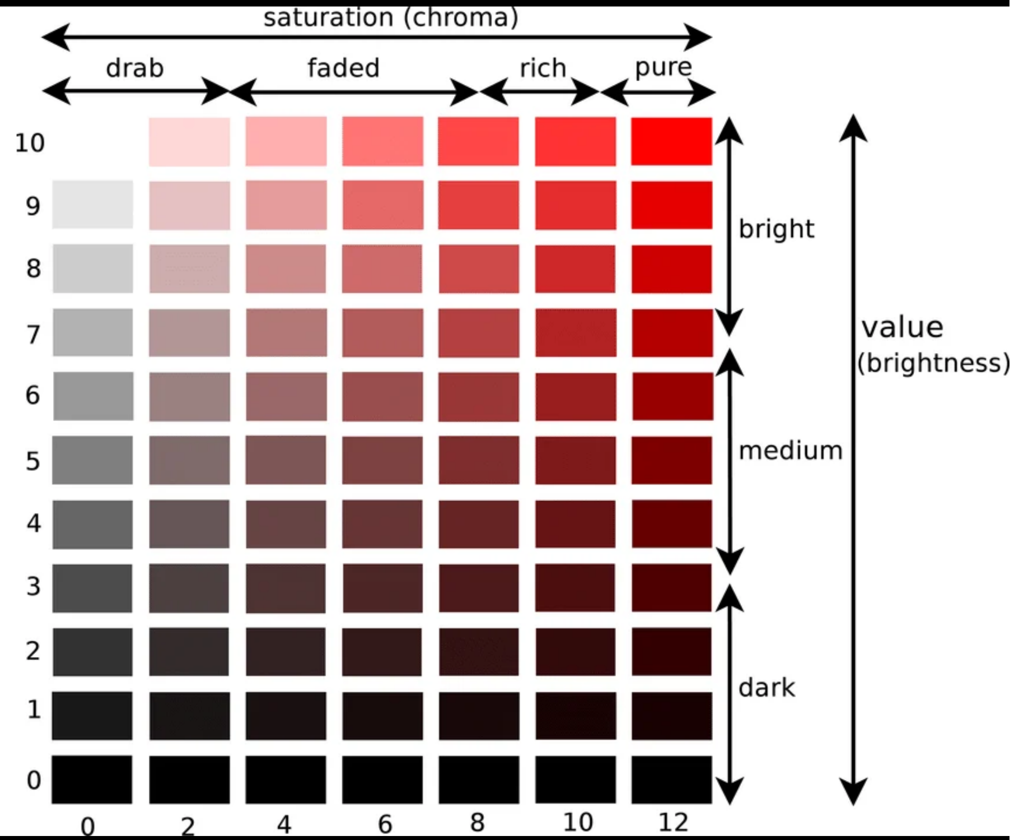
- Common visual variables
  - Shape
  - Size
  - Arrangement
  - Texture
  - orientation
  - Color (hue, value, chroma)

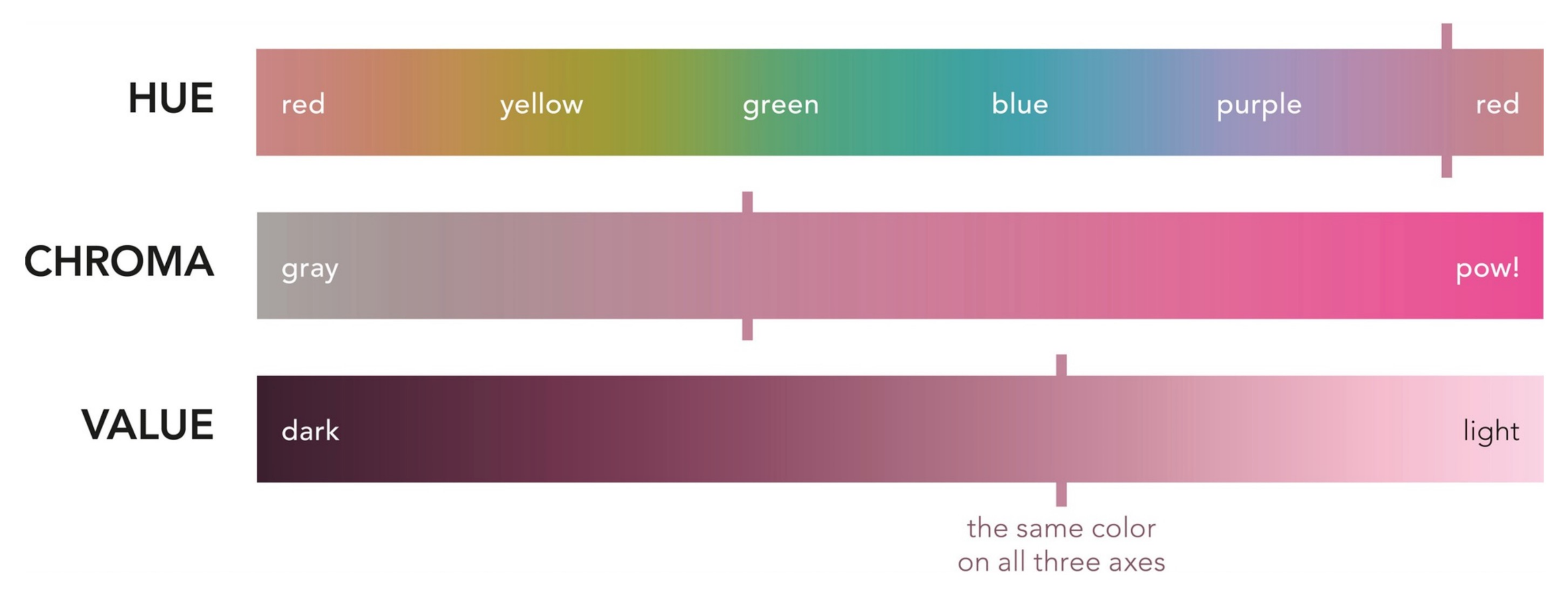


Color

- Hue (色彩 / 色相 / 色度)
  - The appearance of the color, such as red, yellow, blue, etc.
- Value (明度 / 明暗度)
  - The brightness and darkness of the color; that is the amount of light reflected by the color. When there is little reflection, the color is darker and the brightness is low.
- Chroma (彩度)
  - The purity of saturation of a color; that is the amount of solid color contained in a color, and the pure color is the highest saturation among all hues.
- Achromatic (無彩色)
  - White, grey, and black (like greyscale) only have brightness but no hue and saturation.


HUE
YELLOW
primary
YELLOW ORANGE
tertiary
ORANGE
secondary
RED ORANGE
tertiary
RED
primary
RED PURPLE
tertiary
PURPLE
secondary
BLUE PURPLE
tertiary
BLUE
primary
BLUE GREEN
tertiary
GREEN
secondary
YELLOW GREEN
tertiary


saturation (chroma)
drab
faded
rich
pure
10
9
8
7
6
5
4
3
2
1
0
0
2
4
6
8
10
12
bright
medium
dark
value
(brightness)



The three properties of color. Each of these axes is independent of the others, meaning that a color can become grayer without becoming lighter, bluer without becoming darker, and so on. Notice, for example, that all the hues on the top axis have the same chroma (a bit grayish) and value (relatively light); on the second axis the grayest pink on the left is the same lightness as the flaming magenta on the right (try squinting to test it); the last axis keeps constant hue and chroma. (Also notice that hue circles back on itself; the others do not.) Color has been recognized as having three dimensions since the eighteenth century, but chroma—also known as saturation, intensity, or purity—was not fully systematized until Albert Munsell in the early twentieth century.

Color Systems

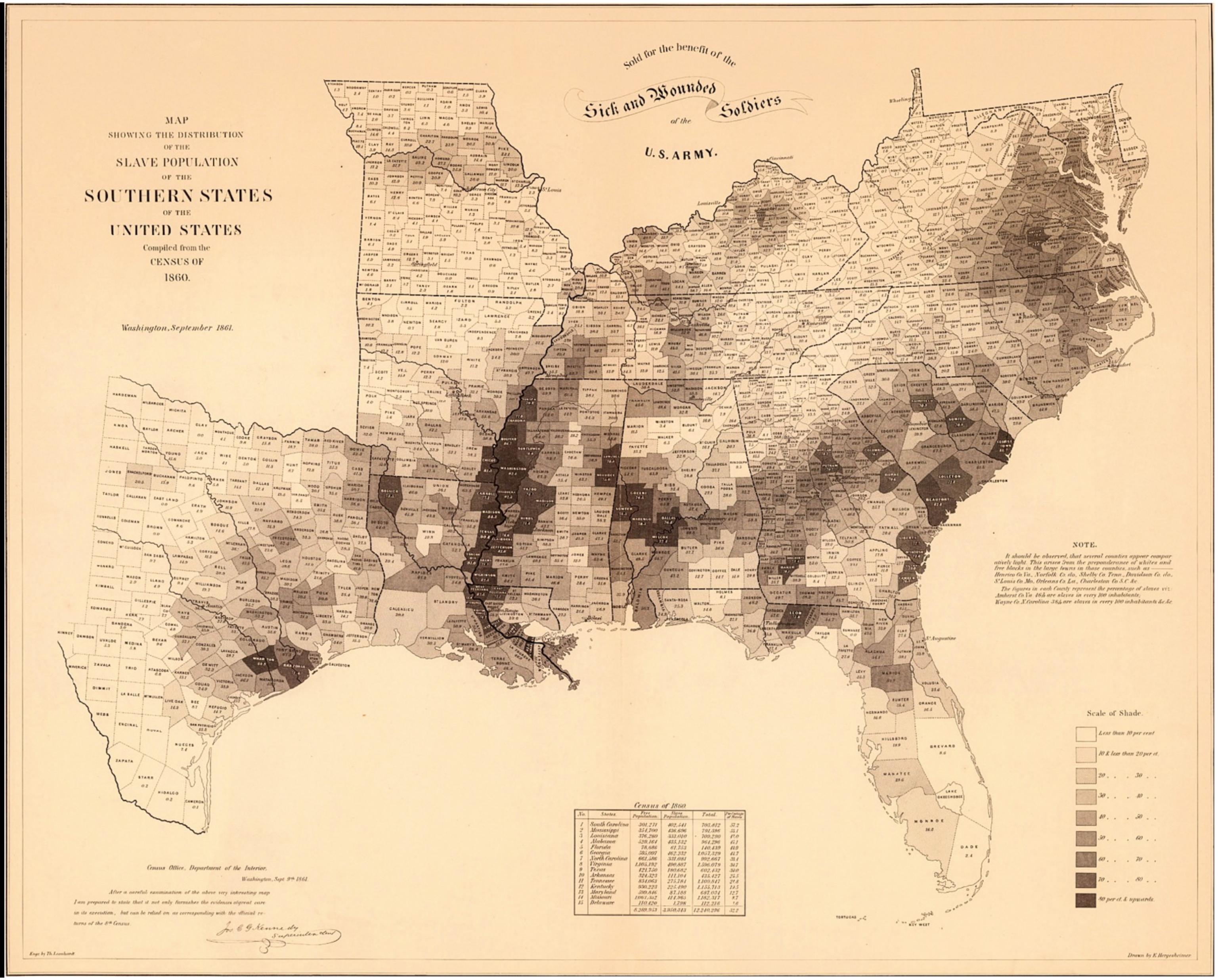
- CIE (International Commission of Illumination, 國際照明委員會)
  - Three primary colors: red, green, and blue are the three colors as the standard, and any color can be mixed with these three primary colors.
- RGB: red, green, and blue
- RGBA: red, green, blue, and alpha (opacity, 不透明度)
- CMYK: cyan (青綠色), magenta (洋紅色), yellow, and black (for print)
- HVC: hue, value, chroma
- HLS: hue, lightness, saturation
- Munsell: red, yellow, green, blue, and purple

Application of Symbols

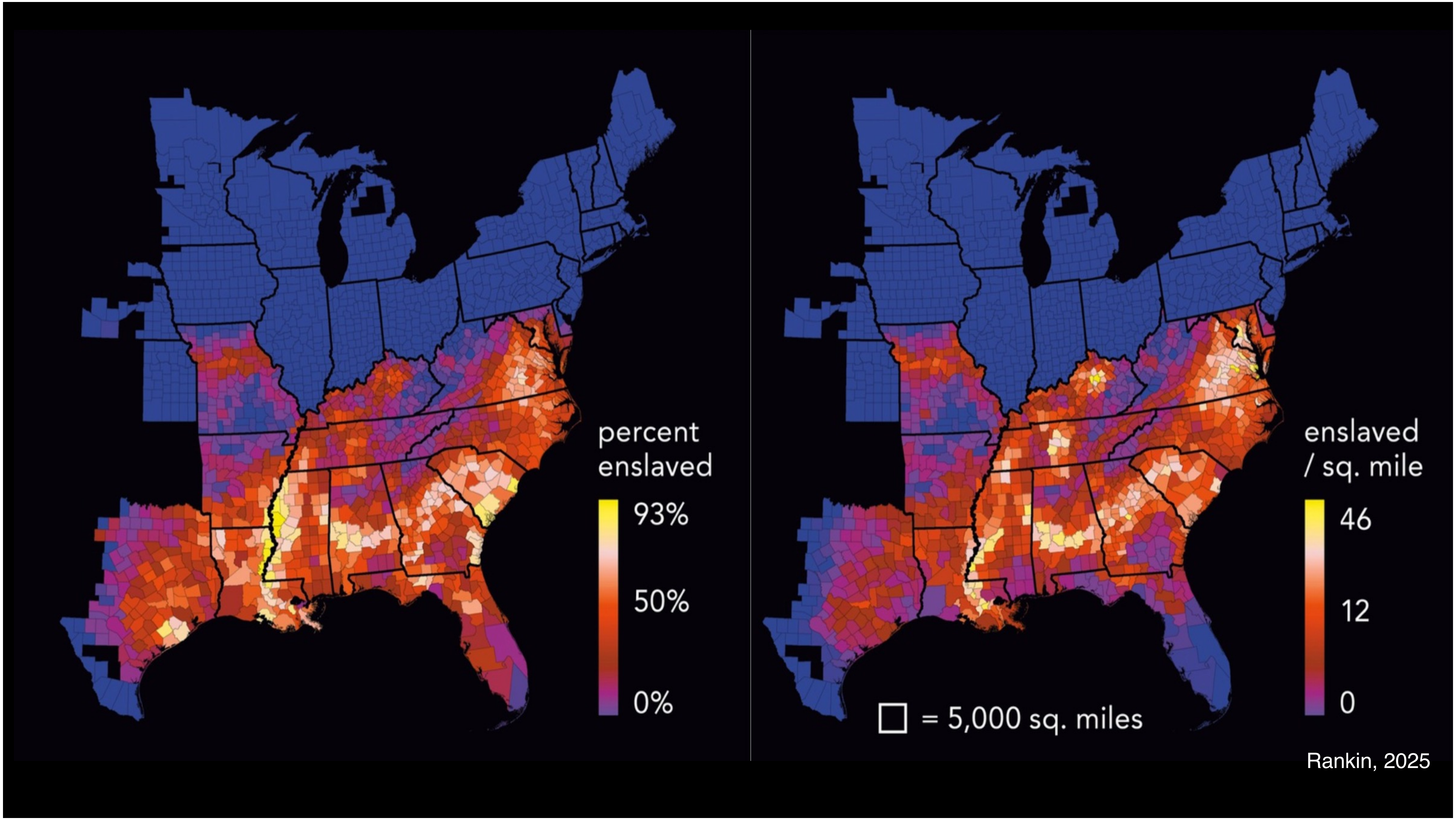
- Value, size, and chroma are commonly used to represent differences in magnitude or level, and are therefore also referred to as ordering variables. They often applied to data measured on ordinal and interval/ratio scale.

- In contrast, combinations of hue, orientation, and shape are used to distinguish differences in categories, and are thus called differentiating variables. There are typically applied to data measured on a nominal scale.

First Reading of the Emancipation Proclamation of President Lincoln, 1862, Francis Bicknell Carpenter

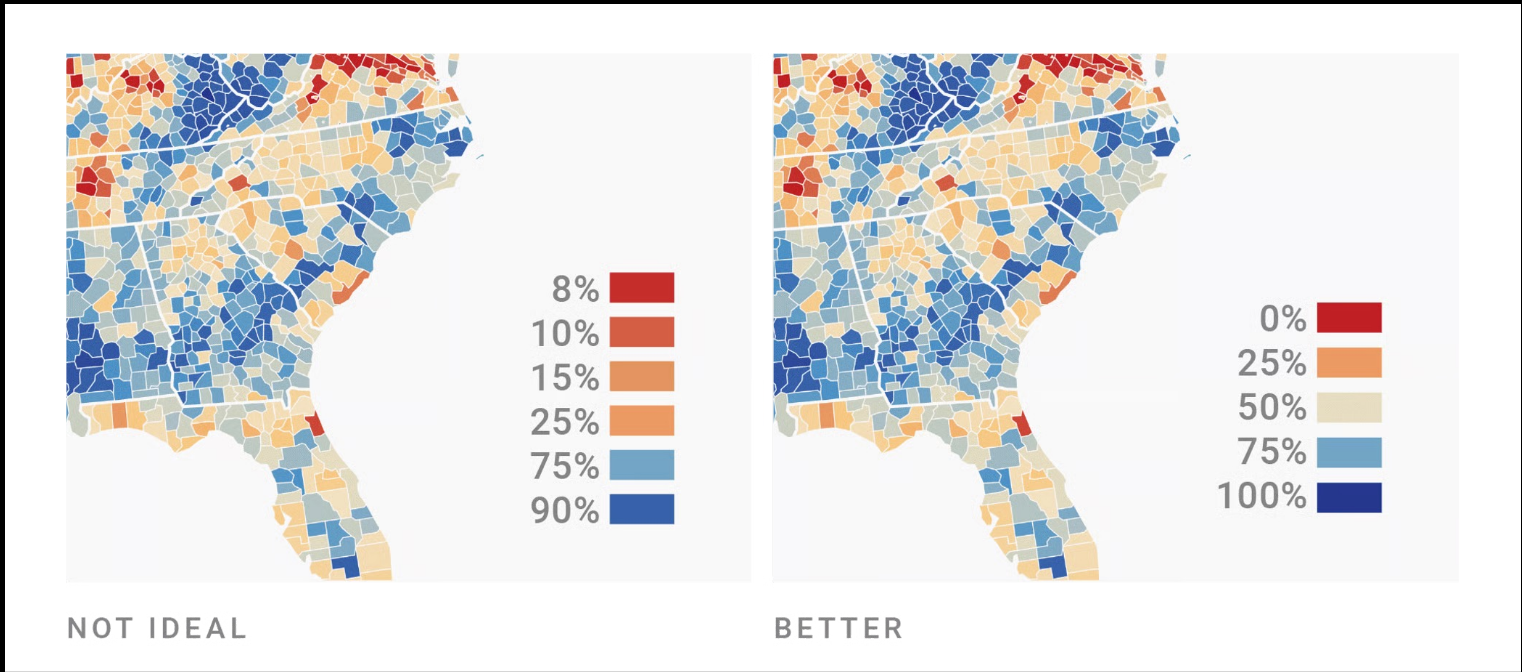


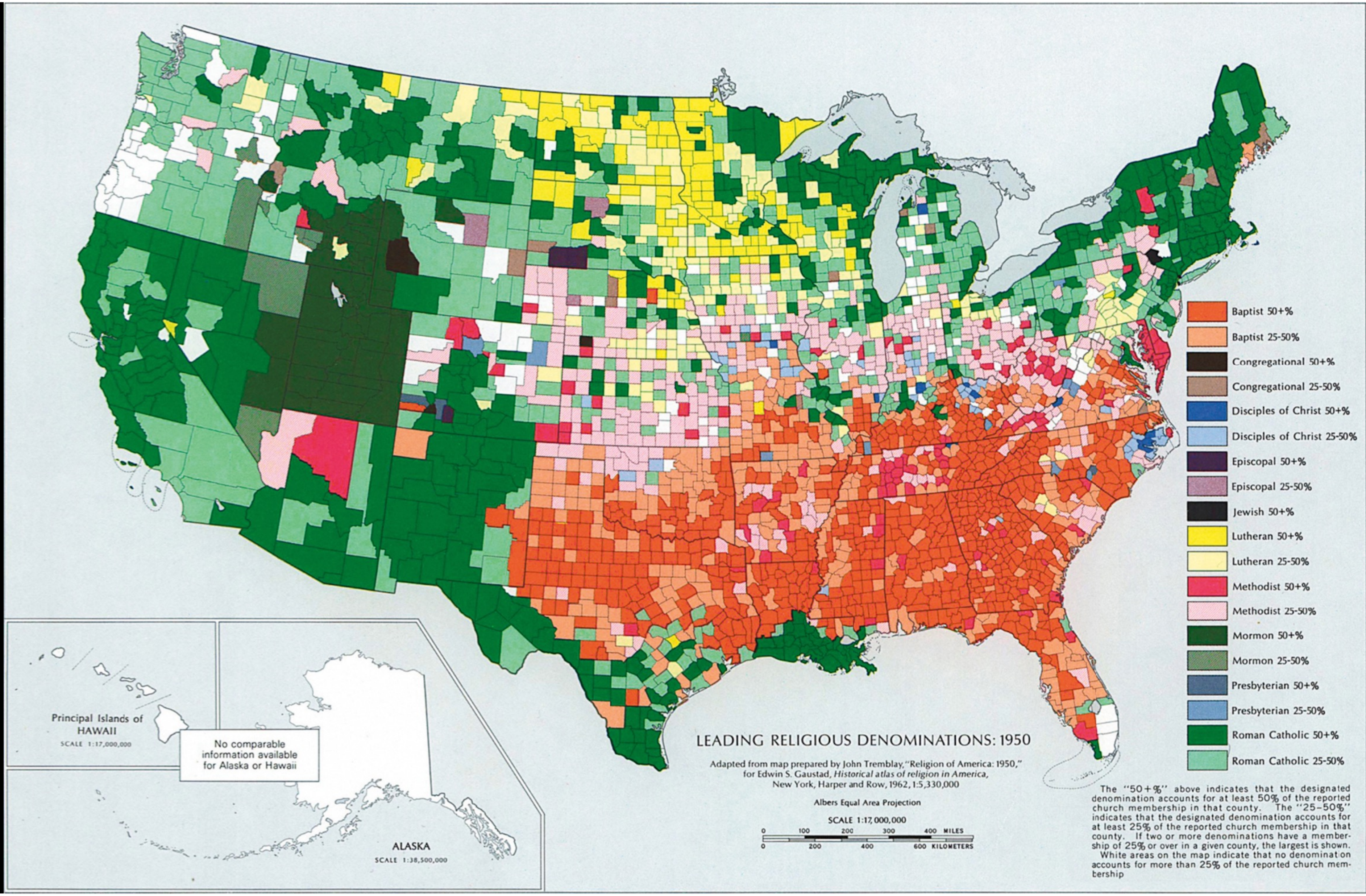
“Lincoln’s map of slavery”,
1861, William Ricketts Palmer


percent
enslaved
93%
50%
0%
enslaved
/ sq. mile
46
12
0
= 5,000 sq. miles
Rankin, 2025

Tips for Map Colors

- Spend time thinking about your color key.





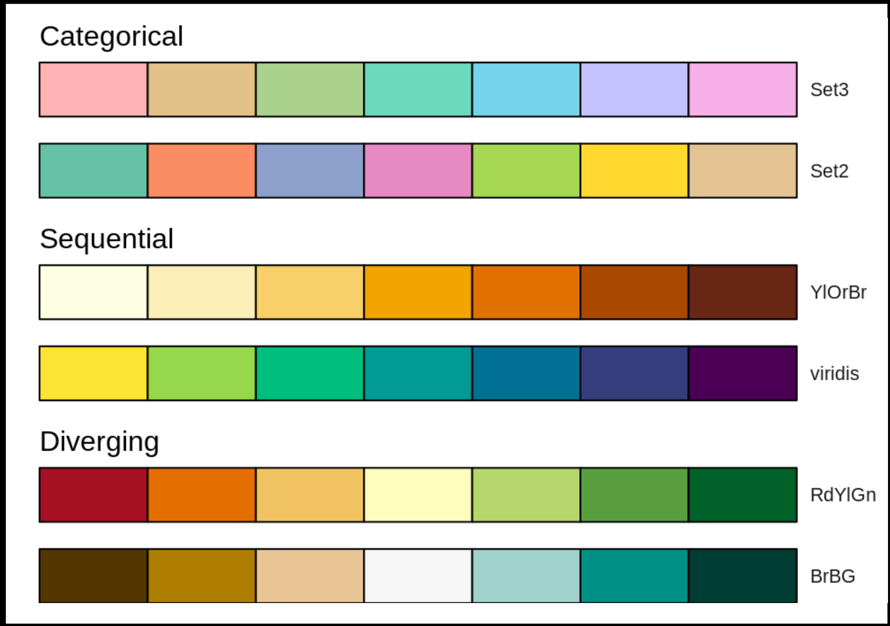
LEADING RELIGIOUS DENOMINATIONS: 1950

Adapted from map prepared by John Tremblay, "Religion of America: 1950," for Edwin S. Gaustad, *Historical atlas of religion in America*, New York, Harper and Row, 1962, 1:5,330,000

The "50+%" above indicates that the designated denomination accounts for at least 50% of the reported church membership in that county. The "25–50%" indicates that the designated denomination accounts for at least 25% of the reported church membership in that county. If two or more denominations have a membership of 25% or over in a given county, the largest is shown.
White areas on the map indicate that no denomination accounts for more than 25% of the reported church membership

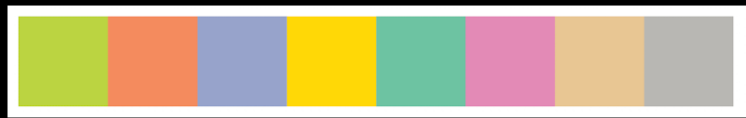
Color Schemes (配色)

- Sequential
  - From bright blue to dark blue
  - e.g. salary
- Diverging
  - From red via white to blue
  - Encode two different ranges of a measure (positive and negative) or a range of a measure between two categories.
  - e.g. temperatures, elevation
- Qualitative/categorical
  - Different color hues to distinguish between different categories.
  - e.g. one green color, one blue color

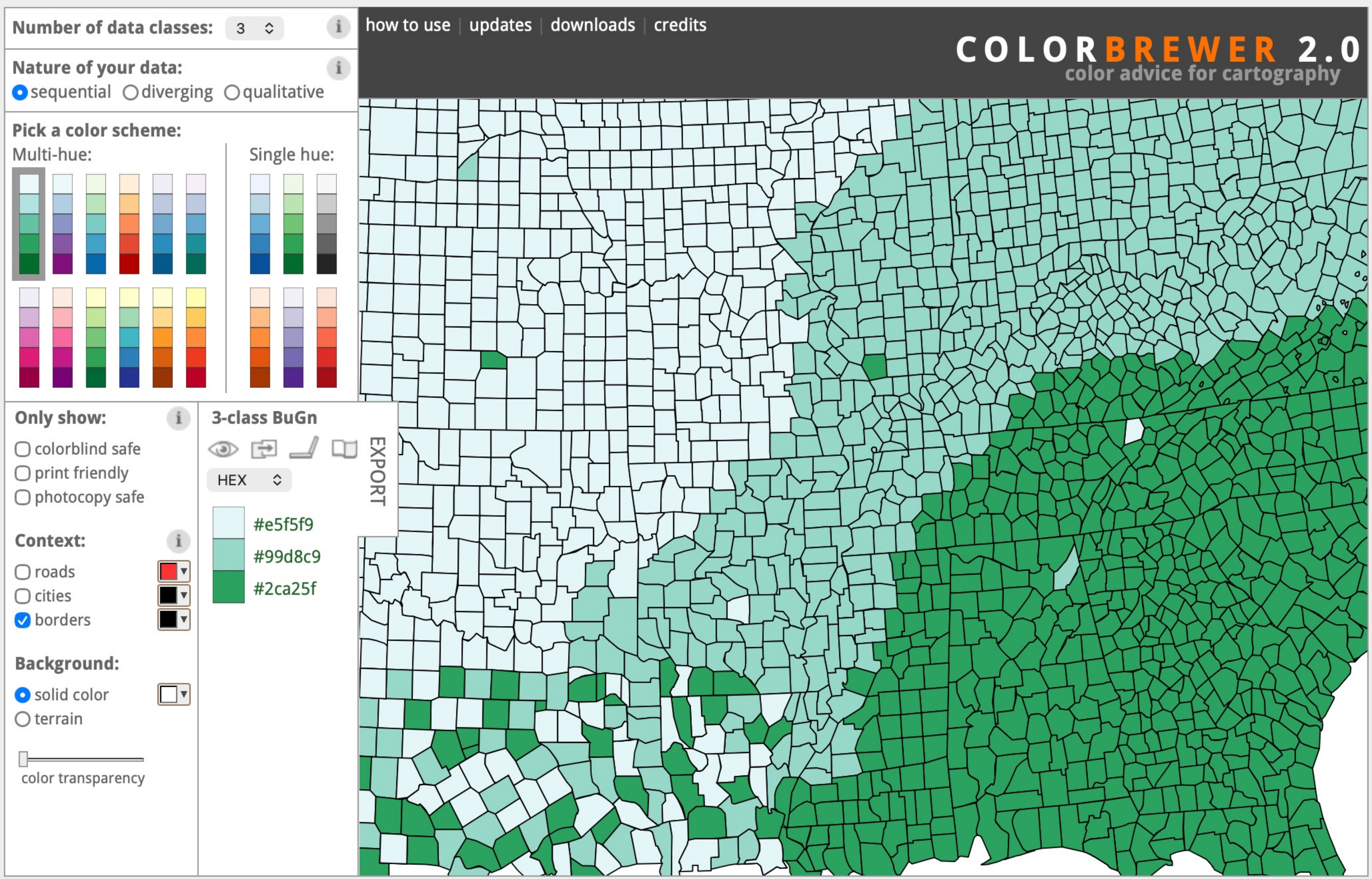


Tips for Map Colors

- When using color to represent categories—different religions, for instance—the standard recommendation is to assign each category a hue of roughly the same value. This group of colors can be shifted lighter or darker as needed, and additional colors can be added for particularly complex maps. Colors are spaced out relatively evenly, with a balance between chromatic and linguistic separation.

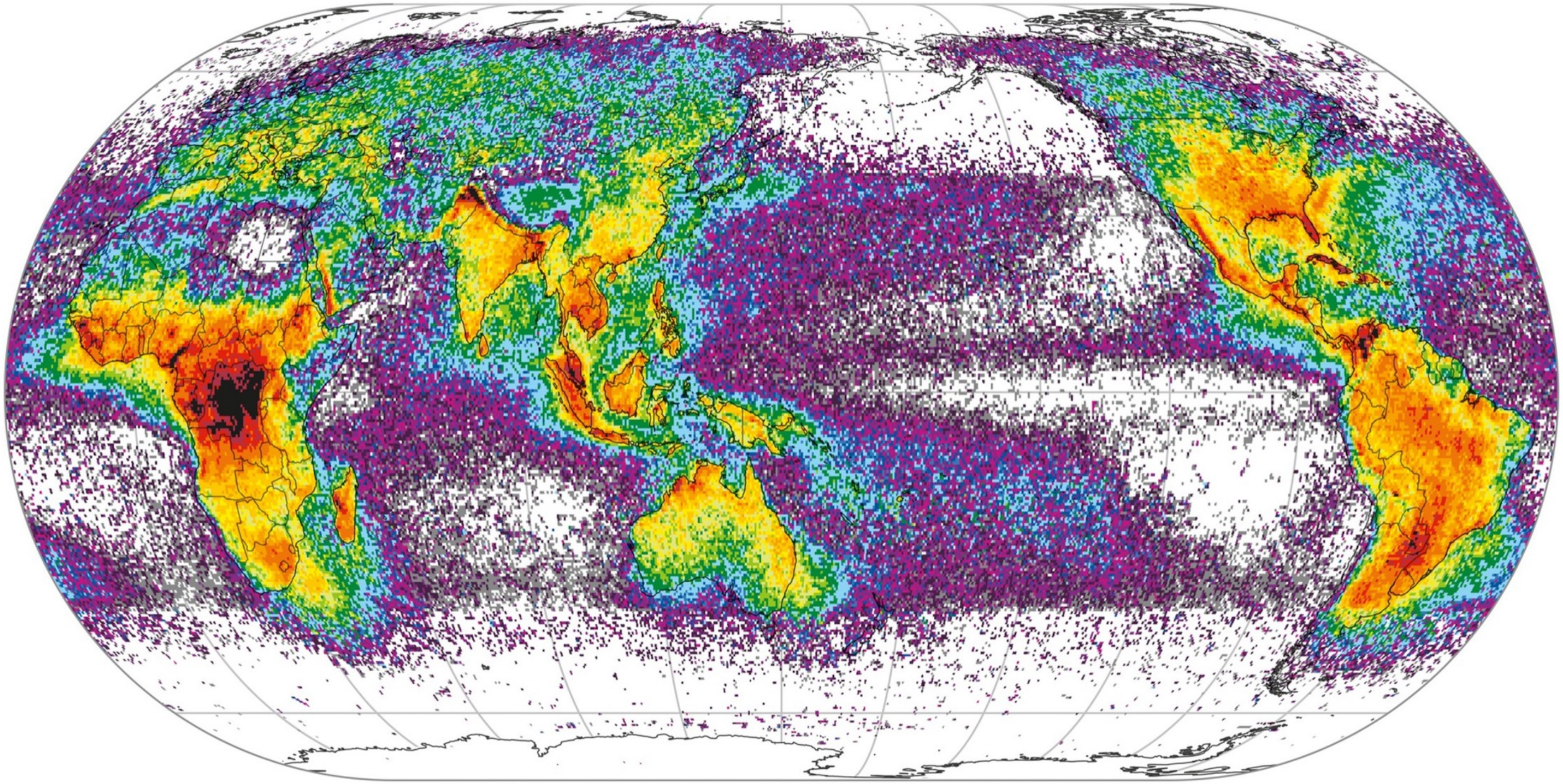


- The idea is to give “the same perceptual weight to each category so that no group is perceived to be larger or more important than any other one.” The colors, in other words, don’t mean anything.

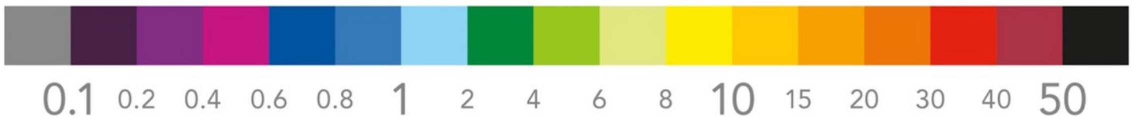

Number of data classes: 3
Nature of your data:
sequential diverging qualitative
Pick a color scheme:
Multi-hue:
Single hue:
Only show:
colorblind safe
print friendly
photocopy safe
Context:
roads
cities
borders
Background:
solid color
terrain
color transparency
3-class BuGn
EXPORT
HEX
#e5f5f9
#99d8c9
#2ca25f
how to use | updates | downloads | credits
COLORBREWER 2.0
color advice for cartography

Let's see more examples!

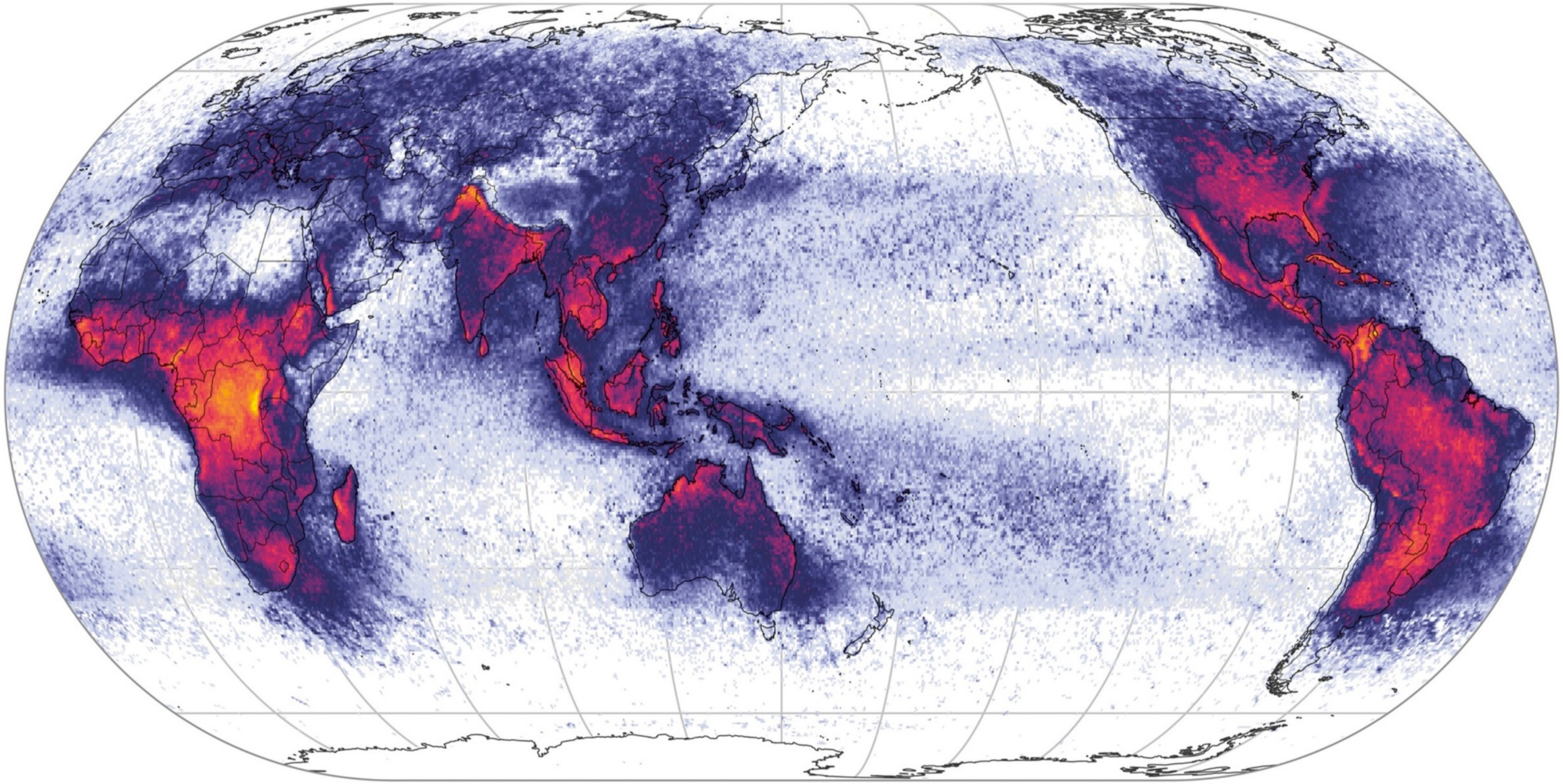
Frequency of Lightning Strikes



colors show number of strikes per square kilometer per year:

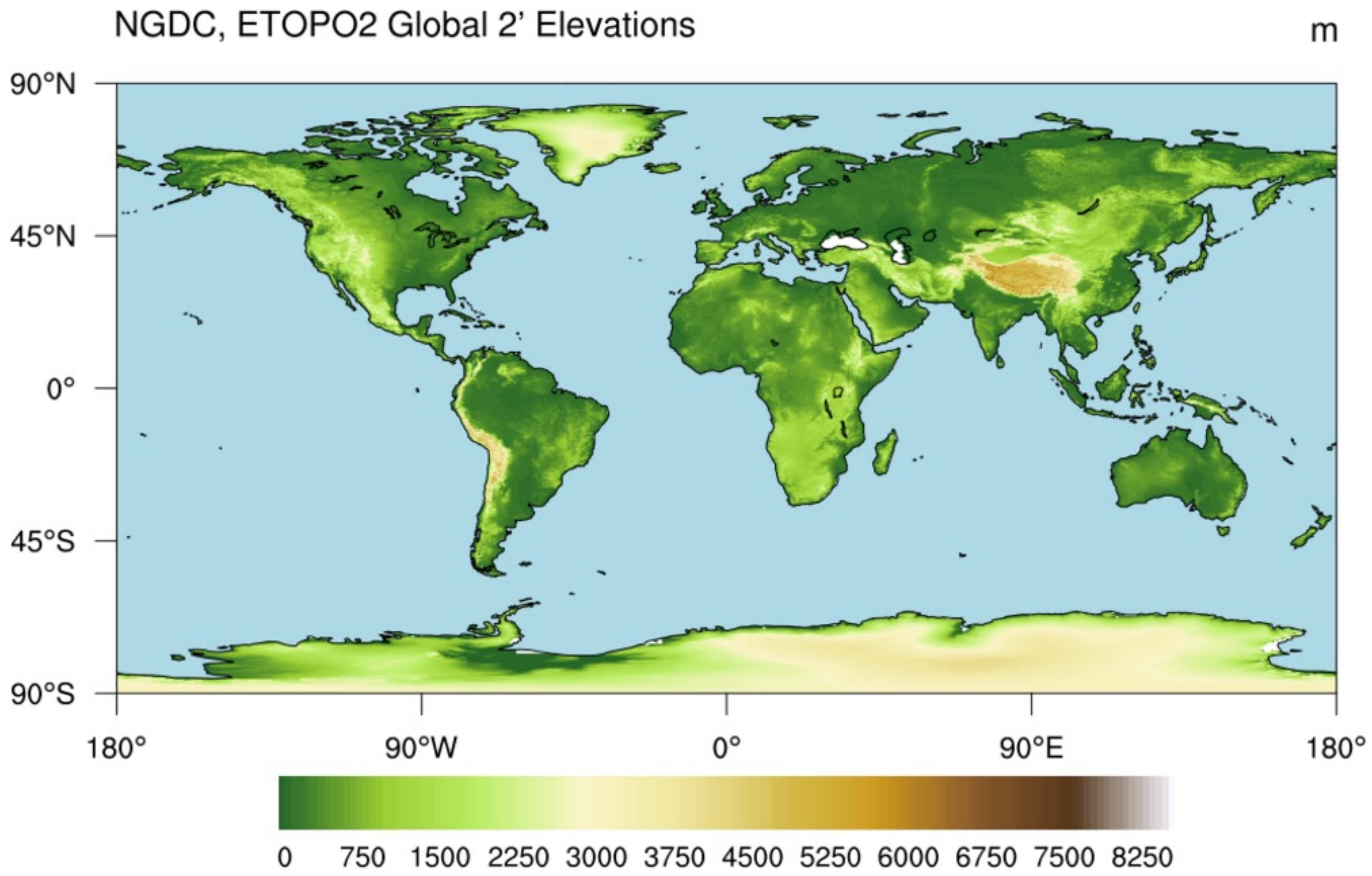


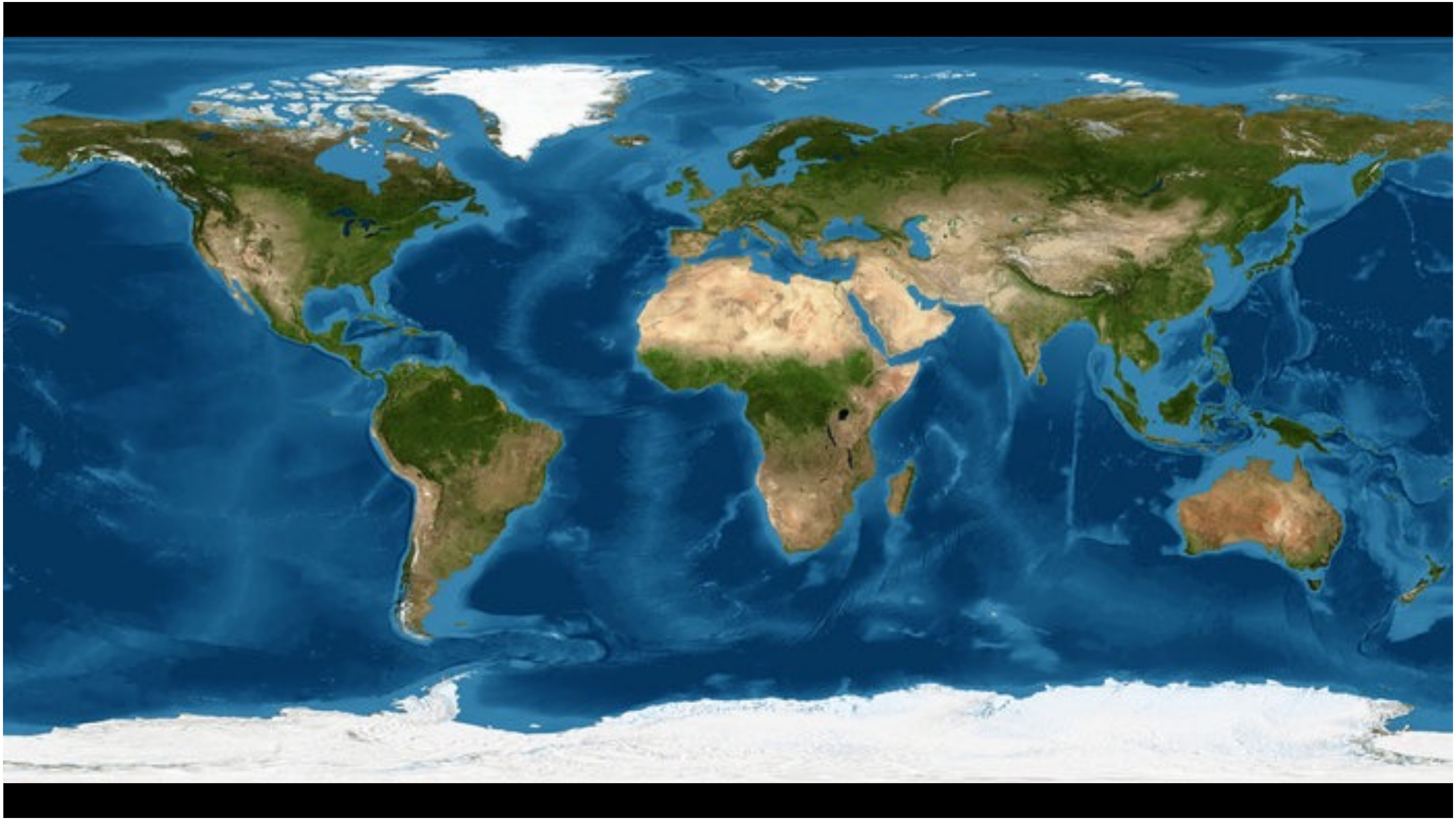
Frequency of Lightning Strikes



colors show number of strikes per square kilometer per year:

0.1 0.2 0.5 1 2 5 10 20 50 100 200

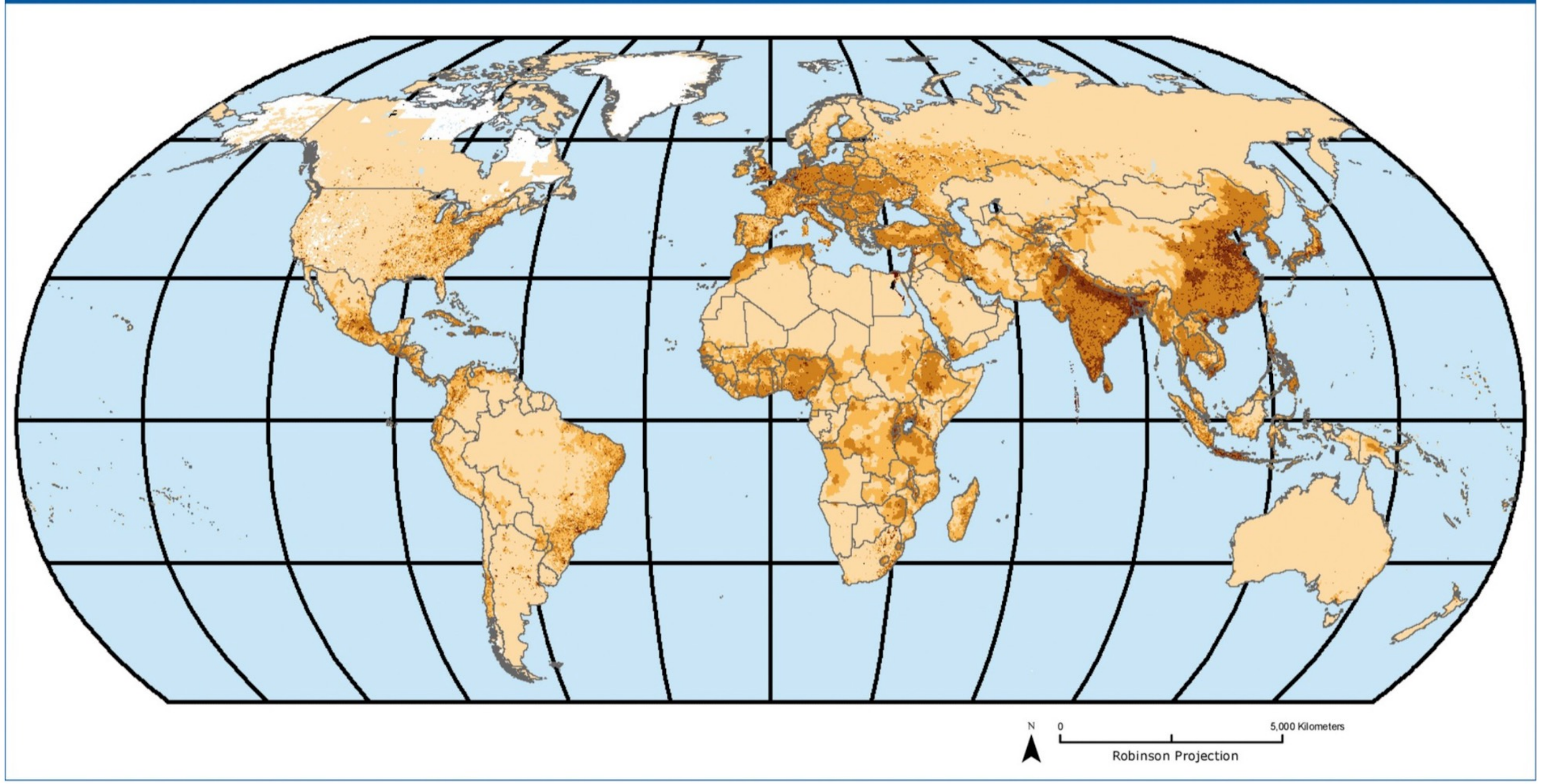

NGDC, ETOPO2 Global 2' Elevations
m
90°N
45°N
0°
45°S
90°S
180°
90°W
0°
90°E
180°
0
750
1500
2250
3000
3750
4500
5250
6000
6750
7500
8250



POPULATION DENSITY, 2000

Global

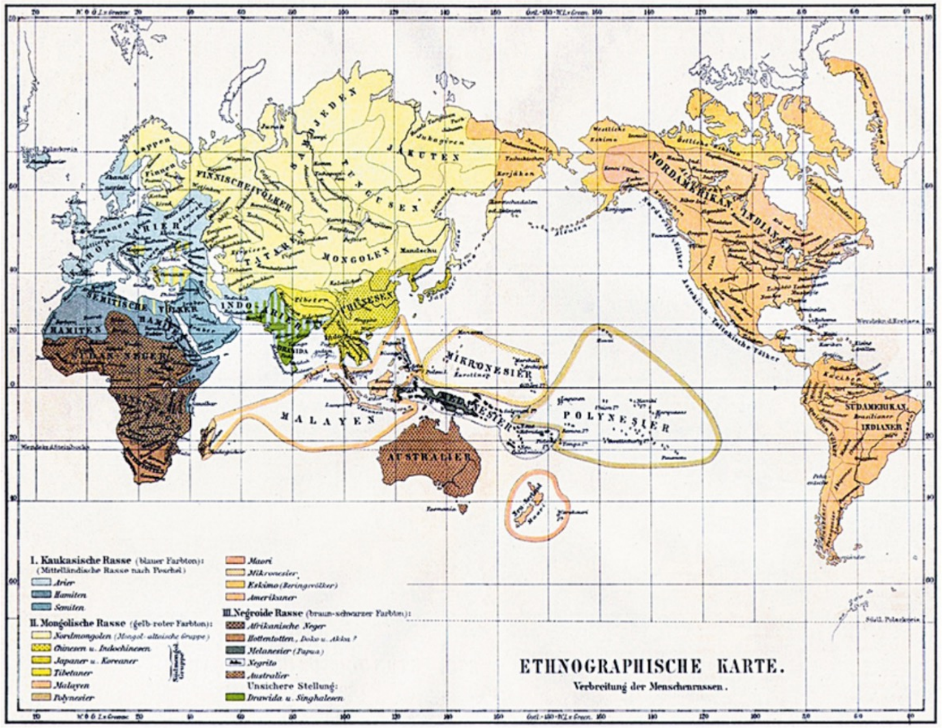
GRUMP v1

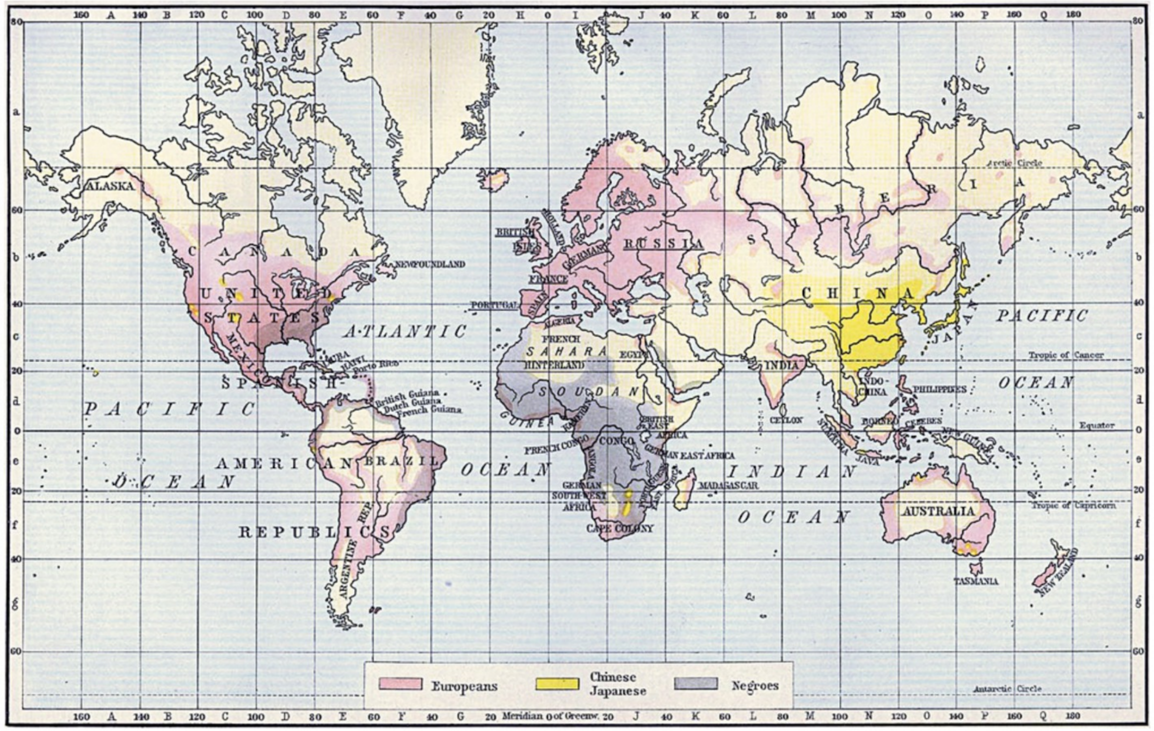


Global Rural–Urban Mapping Project

Persons per km²	Boundaries
0	—— Country
1 - 5	
6 - 25	
26 - 250	
251 - 1,000	
1,001 +	

Population density measures the number of persons per square kilometer of land area. The data are gridded at a resolution of 30 arc-seconds.

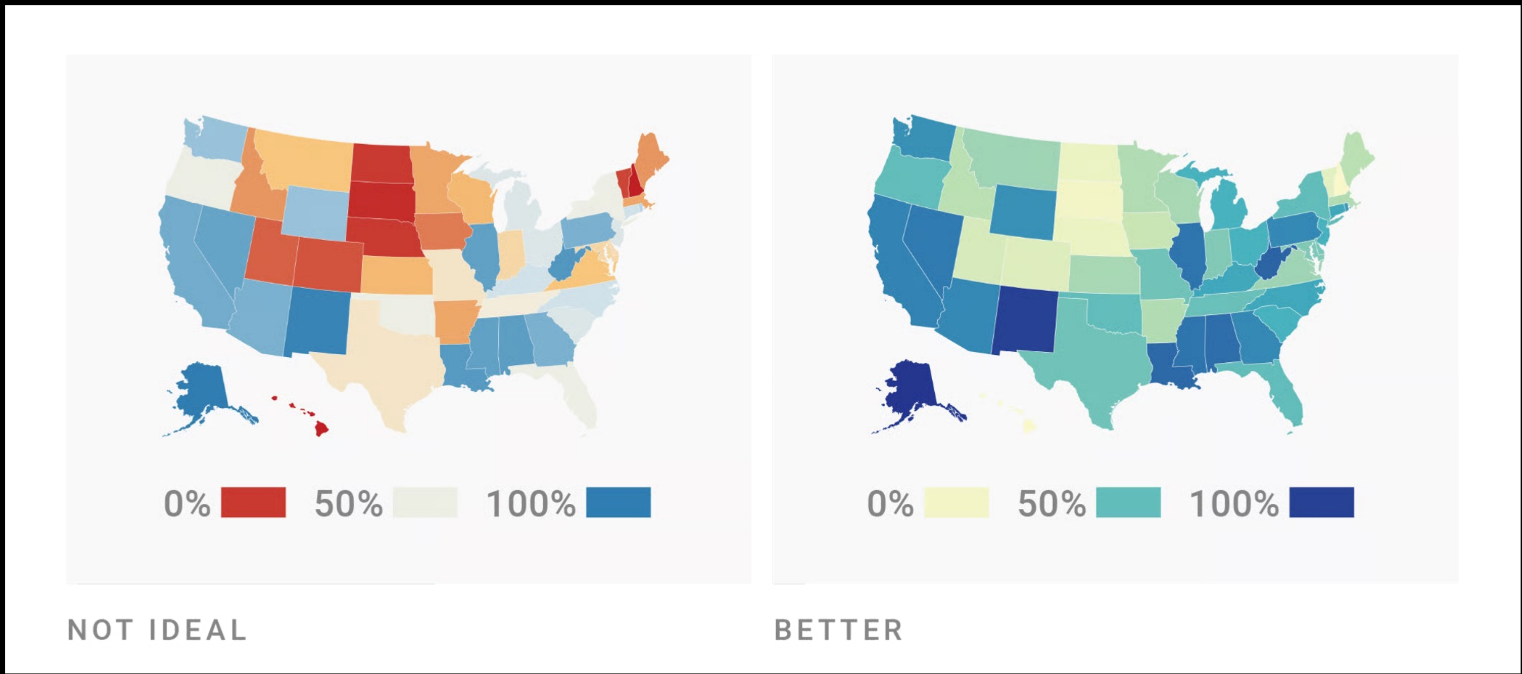




Two maps of racial geography, from an 1888 German encyclopedia (left) and a 1911 American school atlas (right).

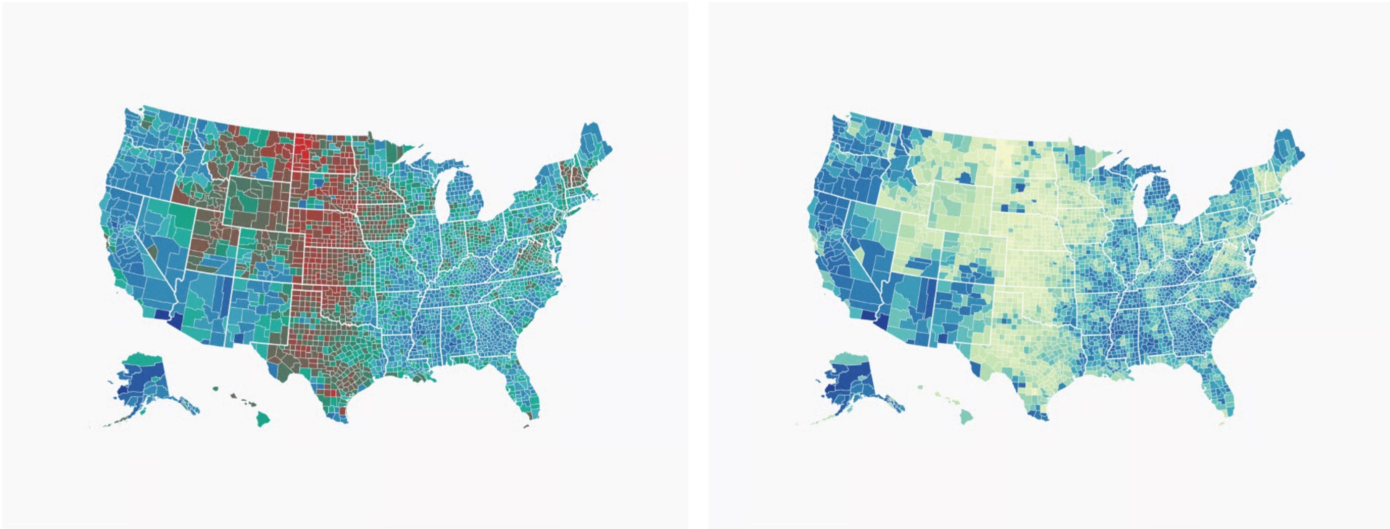
Tips for Map Colors

- Make sure to use the right color scheme for your data, and be aware of natural colors.



Tips for Map Colors

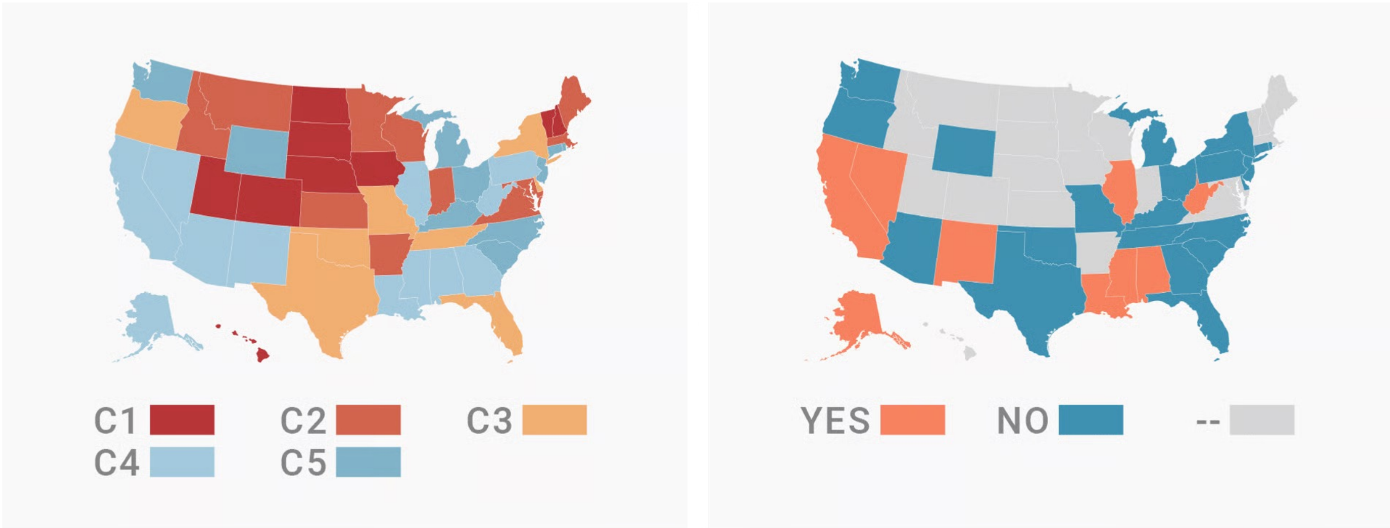
- Make sure there is a lightness difference in your sequential/diverging color schemes.



NOT IDEAL　　　　　　　　　　　　BETTER

Tips for Map Colors

- Consider using as few as possible colors in your qualitative color schemes.

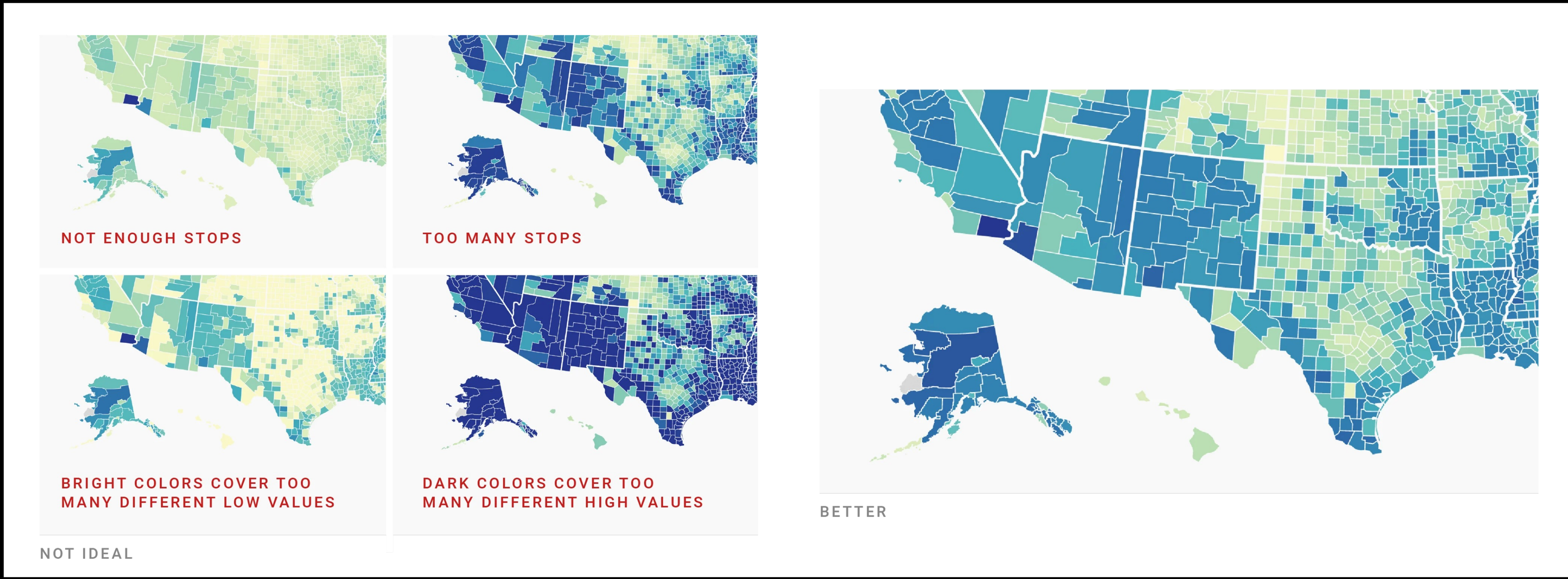


NOT IDEAL

BETTER

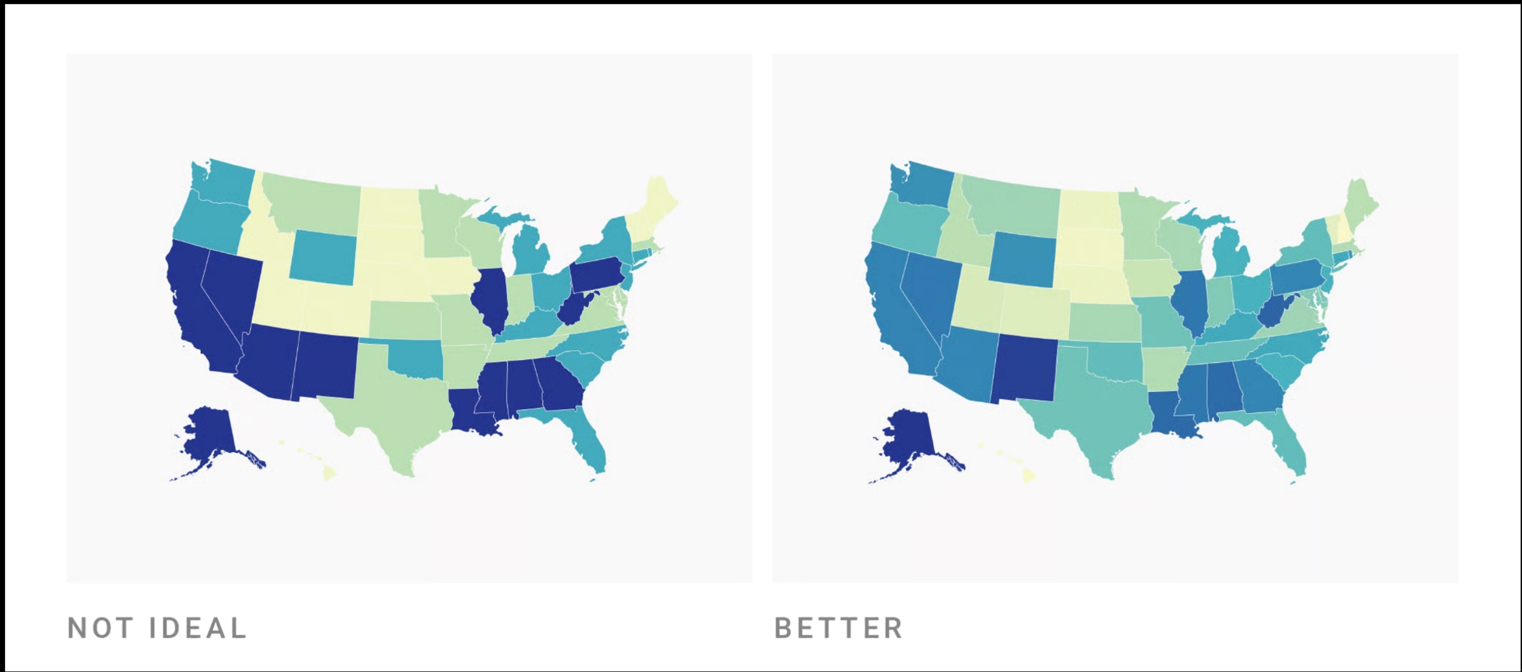
Tips for Map Colors

- Make sure that readers see all the differences in the data.



Tips for Map Colors

- Consider using a continuous color scale instead of discrete color steps.

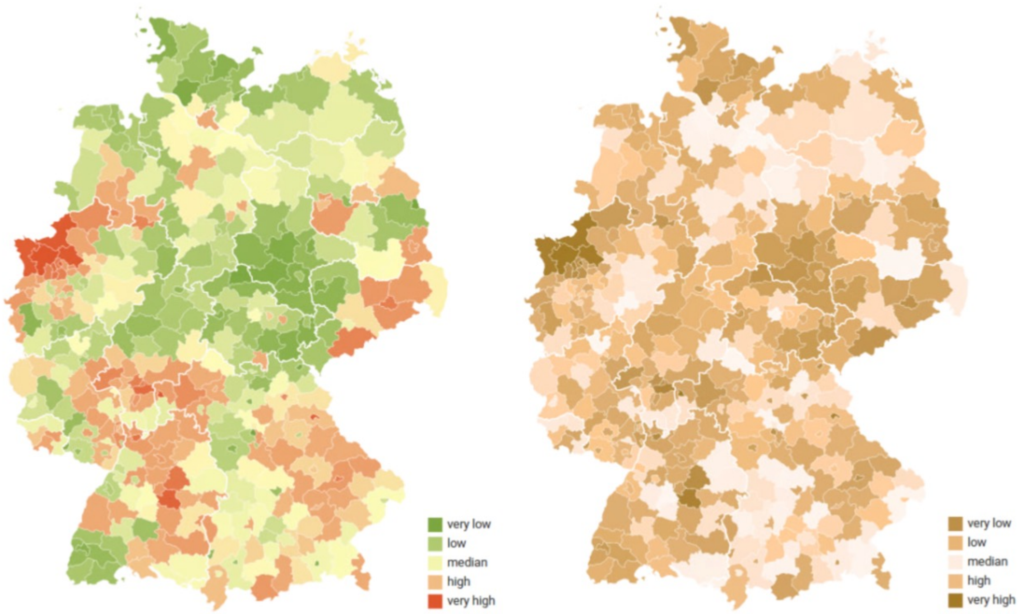


Tips for Map Colors

- Specific categorical color schemes may be used in professional purposes.

Tips for Map Colors

- With any color scheme, do use colorblind-friendly colors.



Left: the map under normal vision, Right: how the map looks for a color blind person (red-green blindness).

[redacted]